

# Recent Progress

Boundary and Nonboundary Localization Quotients of Smooth Affine  $\mathfrak{sl}_2$ -Modules

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## Abstract

We report recent progress on localization quotients of smooth modules over the affine Lie algebra  $\mathfrak{g} = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K$ . The established results are as follows. (1) The boundary isomorphism  $\mathcal{T}_{f_b} M_{a,b}(\chi) \cong M_{a+1,b-1}(\chi^+)$  under  $\lambda_L \neq 0$ , together with its iteration and its compatibility with the degree derivation. (2) For  $c < b$ , the quotient  $Q_c = \mathcal{T}_{f_c} M_{a,b}(\chi)$  has a PBW basis in the negative powers of  $f_c$ ;  $f_c$  acts surjectively and locally nilpotently,  $f_j$  acts injectively for  $j \leq b$ ,  $j \neq c$ , and locally nilpotently for  $j > b$ ; the elements  $h_i$  with  $1 \leq i \leq b-c$  have no nonzero locally finite vectors, and  $h_i - \lambda_i$  is locally nilpotent for  $i > b-c$ ; the annihilator is preserved, and quotient localizations in distinct  $f_j$  commute. (3) Under  $\lambda_L \neq 0$ ,  $\kappa \neq -2$  and  $c \leq -a-1$ , the quotient  $Q_c$  is simple if and only if  $\lambda_0 - c\kappa \notin \mathbb{Z}$ ; for  $\lambda_0 - c\kappa \in \mathbb{Z}$  the subspace  $G_c = \{v : E^n v = 0 \text{ for some } n\}$  is the unique maximal submodule and  $Q_c/G_c$  is simple. (4) Algebraic central reduction, including the Sugawara operators at the critical level, commutes with the quotient localization. For the subspace  $G_c$  we record the decomposition into irreducible modules of the subalgebra generated by  $E = e_{-c}$ ,  $F = f_c$ ,  $H = h_0 - cK$ , the corrected commutation formula for the action of  $E$ , the triangular system determining the vectors annihilated by  $E$ , and a projection lowering the highest weight by 2; the classification of the submodules of  $G_c$ , and the simplicity in the case  $-a \leq c < b$ , remain open.

## 1 Setup and notation

Let  $\mathfrak{g} = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K$  and  $U = U(\mathfrak{g})$ ; write  $x_i = x \otimes t^i$ . For  $i, j \in \mathbb{Z}$ ,

$$\begin{aligned} [e_i, f_j] &= h_{i+j} + i\delta_{i+j,0}K, & [h_i, h_j] &= 2i\delta_{i+j,0}K, \\ [h_i, e_j] &= 2e_{i+j}, & [h_i, f_j] &= -2f_{i+j}, \end{aligned} \tag{1}$$

$K$  is central and  $[e_i, e_j] = [f_i, f_j] = 0$ . Fix  $a, b \in \mathbb{Z}$  with  $a+b \geq 0$ , put  $L = a+b+1 \geq 1$ , and let

$$S_{a,b} = \text{span}_{\mathbb{C}}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

A character  $\chi : S_{a,b} \rightarrow \mathbb{C}$  is given by  $\kappa = \chi(K)$  and  $\lambda_i = \chi(h_i)$ ,  $0 \leq i \leq L$ , with  $\lambda_i := 0$  for  $i > L$  and  $\chi = 0$  on all root vectors of  $S_{a,b}$ . Set

$$M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_{\chi}, \quad u = 1 \otimes 1_{\chi}.$$

We use the criterion of Futorny–Guo–Xue–Zhao [1, Theorem 1.1]:

$$M_{a,b}(\chi) \text{ is simple} \iff \lambda_L \neq 0 \text{ and } \kappa \neq -2. \tag{2}$$

For an element  $F = f_c$  we write  $U_F = U[F^{-1}]$  (an Ore localization, since  $\text{ad } F$  is locally nilpotent on  $U$ ),  $D_F V = U_F \otimes_U V$ , and

$$\mathcal{T}_F V = \text{coker}(V \rightarrow D_F V). \tag{3}$$

The Poincaré–Birkhoff–Witt basis with  $F$  placed first makes  $M_{a,b}(\chi)$  free over  $\mathbb{C}[F]$ , so  $F$  acts injectively and  $\mathcal{T}_F M = D_F M / M$ . For  $c < b$  we abbreviate

$$Q_c = \mathcal{T}_{f_c} M_{a,b}(\chi), \quad w_m = f_c^{-m} u + M \quad (m \geq 1).$$

The two cases  $c \leq -a - 1$  (equivalently  $b - c \geq L$ ) and  $-a \leq c < b$  will be treated separately in Sections 4 and 7.

## 2 The boundary isomorphism

### 2.1 The basic case $\mathcal{T}_{f_1} M_{-1,1}(\varphi) \cong M_{0,0}(\psi)$

Fix  $M = M_{-1,1}(\varphi)$ ,  $u = u_\varphi$ ,  $F = f_1$ ,  $T = \mathcal{T}_F M$ ,  $w_m = F^{-m} u + M$ ; write  $\kappa = \varphi(K)$ ,  $\lambda_i = \varphi(h_i)$ , so

$$e_i u = 0 \quad (i \geq 0), \quad f_j u = 0 \quad (j \geq 2), \quad h_i u = \lambda_i u \quad (i \geq 0), \quad Ku = \kappa u, \quad (4)$$

with  $\lambda_i = 0$  for  $i \geq 2$ . Define  $\psi : S_{0,0} \rightarrow \mathbb{C}$  by  $\psi(K) = \kappa$ ,  $\psi(h_0) = \lambda_0 + 2$ ,  $\psi(h_1) = \lambda_1$ ,  $\psi(h_i) = 0$  ( $i \geq 2$ ),  $\psi(e_i) = \psi(f_i) = 0$  ( $i \geq 1$ ), and put  $A = M_{0,0}(\psi)$ ,  $v = 1 \otimes 1_\psi$ .

For a finite nondecreasing integer sequence  $\alpha = (i_1, \dots, i_k)$  with  $i_\nu \leq r$  (including the empty sequence) write  $\ell(\alpha) = k$ ,  $x(\alpha) = x_{i_1} \cdots x_{i_k}$ ,  $x(\emptyset) = 1$ . Let

$$\mathcal{X} = \{X = e(\alpha)h(\beta)f(\gamma) : \alpha, \beta \in \Gamma_{-1}, \gamma \in \Gamma_0\}, \quad d(X) = 2\ell(\alpha) + \ell(\beta),$$

and set  $B_{m,X} = F^{-m} Xu + M$  and  $T_{\leq q} = \text{span}_{\mathbb{C}}\{B_{m,X} : m \geq 1, d(X) \leq q\}$  ( $T_{\leq q} = 0$  for  $q < 0$ ). Then  $\{F^r Xu : r \geq 0, X \in \mathcal{X}\}$  is a basis of  $M$ ,  $\{F^r Xu : r \in \mathbb{Z}, X \in \mathcal{X}\}$  is a basis of  $D_F M$ , and  $\{B_{m,X} : m \geq 1, X \in \mathcal{X}\}$  is a basis of  $T$ ; the source  $A$  has basis  $\{Xe_0^p v : p \geq 0, X \in \mathcal{X}\}$  with  $e_0$  placed after all factors of  $X$  and before  $S_{0,0}$ . In particular  $w_1 \neq 0$ .

Put  $D = \text{ad } F$ . From (1),

$$\begin{aligned} D(f_n) &= 0, & D(h_n) &= 2f_{n+1}, \\ D(e_n) &= -(h_{n+1} + n\delta_{n+1,0}K), & D^2(e_n) &= -2f_{n+2}, & D^3(e_n) &= 0. \end{aligned} \quad (5)$$

**Lemma 2.1.** *For  $x \in U$  and  $m \geq 1$ , the finite sum*

$$xF^{-m} = \sum_{j \geq 0} \binom{m+j-1}{j} F^{-m-j} D^j(x) \quad (6)$$

*holds in  $U_F$ .*

*Proof.* For  $m = 1$ , multiply the right side by  $F$  on the right and use  $D^j(x)F = FD^j(x) - D^{j+1}(x)$ ; the sum telescopes to  $x$ . Suppose (6) holds for  $m$ . Using  $yF^{-1} = \sum_{k \geq 0} F^{-1-k} D^k(y)$  for  $y = D^j(x)$  gives

$$xF^{-(m+1)} = \sum_{j,k \geq 0} \binom{m+j-1}{j} F^{-m-j-1-k} D^{j+k}(x),$$

whose coefficient of  $F^{-(m+1)-r} D^r(x)$  is  $\sum_{j=0}^r \binom{m+j-1}{j} = \binom{m+r}{r}$ , as required.  $\square$

Combining (6) with (5) gives the commutation formulas

$$\begin{aligned} f_n F^{-m} &= F^{-m} f_n, \\ h_n F^{-m} &= F^{-m} h_n + 2m F^{-m-1} f_{n+1}, \\ e_n F^{-m} &= F^{-m} e_n - m F^{-m-1} (h_{n+1} + n\delta_{n+1,0}K) - m(m+1) F^{-m-2} f_{n+2}. \end{aligned} \quad (7)$$

Applying them to  $u$  with (4) and  $m = 1$  gives

$$\begin{aligned} e_n w_1 &= f_n w_1 = 0 \quad (n \geq 1), & K w_1 &= \kappa w_1, \\ h_0 w_1 &= (\lambda_0 + 2) w_1, & h_1 w_1 &= \lambda_1 w_1, & h_n w_1 &= 0 \quad (n \geq 2), \end{aligned} \quad (8)$$

because  $f_1 w_1 = u + M = 0$ ; the corrections for  $e_n$  vanish by  $h_{n+1} u = f_{n+2} u = 0$  ( $n \geq 1$ ), those for  $h_n$  by  $f_{n+1} u = 0$  ( $n \geq 1$ ), and the correction for  $h_0$  is  $2F^{-2} f_1 u + M = 2w_1$ . Thus  $sw_1 = \psi(s)w_1$  for all  $s \in S_{0,0}$ ; since the unital algebra character  $\psi_U : U(S_{0,0}) \rightarrow \mathbb{C}$  extends  $\psi$ , we get  $aw_1 = \psi_U(a)w_1$  for  $a \in U(S_{0,0})$ . Hence the map

$$\Phi : A \longrightarrow T, \quad x \otimes z 1_\psi \longmapsto z x w_1,$$

kills every balancing relation  $xa \otimes z 1_\psi - x \otimes a(z 1_\psi)$ , and descends to a  $U$ -linear  $\Phi$ . From (7) with  $n = 0$ ,  $e_0 w_m = -m \lambda_1 w_{m+1}$ , hence

$$e_0^p w_1 = (-1)^p p! \lambda_1^p w_{p+1} \quad (p \geq 0). \quad (9)$$

**Lemma 2.2.** *For  $m \geq 1$  and  $X \in \mathcal{X}$ ,*

$$X w_m = B_{m,X} + R_{m,X}, \quad R_{m,X} \in T_{\leq d(X)-1}.$$

*Proof.* Give the generators weights  $\text{wt}(e_i) = 2$ ,  $\text{wt}(h_i) = 1$ ,  $\text{wt}(f_i) = \text{wt}(K) = 0$ , and filter  $U$  by total word weight,  $U_{\leq q} = 0$  for  $q < 0$ . By (1) the weight of a bracket is at most the sum of the input weights: for  $[h, e]$ ,  $[e, f]$ ,  $[h, f]$ ,  $[h, h]$  the output weights are at most 2, 1, 0, 0 against input sums 3, 2, 1, 2. So reordering in the PBW basis never increases the weight, and neither does evaluation of the generators of  $S_{0,0}$  on  $u$ . By (5) and the Leibniz rule,  $D(U_{\leq q}) \subseteq U_{\leq q-1}$ , hence  $D^j(X) \in U_{\leq d(X)-j}$  and  $D^j(X) = 0$  for  $j > d(X)$ . Expanding in the PBW basis of  $M$ ,

$$D^j(X)u = \sum_{r \geq 0, X' \in \mathcal{X}} c_{j,r,X'} F^r X' u, \quad c_{j,r,X'} \neq 0 \Rightarrow d(X') \leq d(X) - j.$$

Substitute into (6). The  $j = 0$  term is exactly  $B_{m,X}$ . For  $j \geq 1$ , a term with  $r \geq m + j$  lies in  $M$  and vanishes in  $T$ ; every surviving term is  $B_{m+j-r,X'}$  with  $d(X') \leq d(X) - j < d(X)$ .  $\square$

**Theorem 2.3.** *If  $\lambda_1 \neq 0$ , then  $\Phi : A \xrightarrow{\sim} T$ , and  $\Phi$  is the unique  $\mathfrak{g}$ -module isomorphism sending  $v$  to  $w_1$ . This holds for every  $\kappa \in \mathbb{C}$ .*

*Proof. Surjectivity.* Prove by induction on  $d \geq 0$  the statement  $P(d)$ :  $B_{m,X} \in \text{Im } \Phi$  for every  $m \geq 1$  and every  $X \in \mathcal{X}$  with  $d(X) \leq d$  (all negative powers simultaneously). For  $d = 0$  we have  $X = f(\gamma)$ , which commutes with  $F$ , so  $B_{m,X} = X w_m = \Phi(X q_m)$  with  $q_m = \frac{(-1)^{m-1}}{(m-1)! \lambda_1^{m-1}} e_0^{m-1} v$ , using (9). If  $d \geq 1$  and  $P(d-1)$  holds, Lemma 2.2 gives  $\Phi(X q_m) = B_{m,X} + \sum_\nu c_\nu B_{m_\nu, X_\nu}$  with  $d(X_\nu) \leq d-1$ ; by induction choose  $a_\nu$  with  $\Phi(a_\nu) = B_{m_\nu, X_\nu}$ , and  $B_{m,X} = \Phi(X q_m - \sum_\nu c_\nu a_\nu)$ . The recursion strictly decreases the nonnegative integer  $d$  and terminates. For example  $d(h_{-1}) = 1$  and  $h_{-1} w_m = B_{m, h_{-1}} + 2m B_{m+1, f_0}$ , giving the preimage  $B_{m, h_{-1}} = \Phi(h_{-1} q_m - 2m f_0 q_{m+1})$ .

*Injectivity.* By (9) and Lemma 2.2,

$$\Phi(X e_0^p v) \equiv (-1)^p p! \lambda_1^p B_{p+1, X} \pmod{T_{\leq d(X)-1}}. \quad (10)$$

Let  $0 \neq a = \sum_{X,p} a_{X,p} X e_0^p v \in A$  and  $d = \max\{d(X) : a_{X,p} \neq 0\}$ . In  $T_{\leq d}/T_{\leq d-1}$ ,

$$\Phi(a) + T_{\leq d-1} = \sum_{\substack{X,p \\ d(X)=d}} a_{X,p} (-1)^p p! \lambda_1^p (B_{p+1, X} + T_{\leq d-1}),$$

and these cosets are linearly independent; since  $(-1)^p p! \lambda_1^p \neq 0$ ,  $\Phi(a) \neq 0$ .  $\square$

## 2.2 The general boundary and iteration

**Theorem 2.4.** *Let  $N \geq 1$ ,  $M = M_{-1,N}(\varphi)$ ,  $F = f_N$ ,  $w_m = F^{-m}u + M$ , and assume  $\lambda_N \neq 0$ . Define  $\psi : S_{0,N-1} \rightarrow \mathbb{C}$  by  $\psi(K) = \kappa$ ,  $\psi(h_i) = \lambda_i + 2\delta_{i,0}$  ( $i \geq 0$ ),  $\psi(e_i) = 0$  ( $i \geq 1$ ),  $\psi(f_j) = 0$  ( $j \geq N$ ). Then*

$$\Phi_N : M_{0,N-1}(\psi) \xrightarrow{\sim} \mathcal{T}_{f_N}M, \quad v \mapsto f_N^{-1}u + M.$$

*Proof.* In (5)–(7) replace  $n+1, n+2$  by  $n+N, n+2N$ . Since  $e_i u = 0$  ( $i \geq 0$ ),  $f_j u = 0$  ( $j > N$ ) and  $\lambda_i = 0$  ( $i > N$ ), we get  $e_i w_1 = 0$  ( $i \geq 1$ ),  $f_j w_1 = 0$  ( $j \geq N$ ) (with  $f_N w_1 = u + M = 0$ ) and  $h_i w_1 = (\lambda_i + 2\delta_{i,0})w_1$ , so the balancing argument gives a well-defined  $\Phi_N$ . As  $e_0 u = f_{2N} u = 0$  and  $h_N u = \lambda_N u$ ,

$$e_0 w_m = -m\lambda_N w_{m+1}, \quad w_m = \Phi_N \left( \frac{(-1)^{m-1} e_0^{m-1} v}{(m-1)! \lambda_N^{m-1}} \right).$$

Replace  $\mathcal{X}$  by  $\mathcal{X}_N = \{e(\alpha)h(\beta)f(\gamma) : \alpha, \beta \in \Gamma_{-1}, \gamma \in \Gamma_{N-1}\}$ ; the source and target bases are  $\{X e_0^p v\}$  and  $\{F^{-m} X u + M\}$ , and the same degree  $d(X) = 2\ell(\alpha) + \ell(\beta)$  is strictly lowered by  $\text{ad } f_N$  while reordering and the relations in the inducing subalgebra do not increase it. The same strong induction proves surjectivity, and the coefficients  $(-1)^p p! \lambda_N^p \neq 0$  prove injectivity.  $\square$

**Corollary 2.5.** *Let  $a, b \in \mathbb{Z}$ ,  $a + b \geq 0$ ,  $L = a + b + 1$ ,  $M = M_{a,b}(\chi)$ . If  $\lambda_L \neq 0$ , then*

$$M_{a+1,b-1}(\chi^+) \xrightarrow{\sim} \mathcal{T}_{f_b}M, \quad v \mapsto f_b^{-1}u + M,$$

where  $\chi^+$  vanishes on all root vectors of  $S_{a+1,b-1}$  and  $\chi^+(K) = \kappa$ ,  $\chi^+(h_i) = \lambda_i + 2\delta_{i,0}$ .

*Proof.* Set  $F = f_b$ ,  $Y = e_{a+1}$ ,  $w_m = F^{-m}u + M$ . For  $i \geq a+2$ , commuting  $e_i$  past  $F^{-1}$  produces the Cartan index  $i + b \geq L + 1$  and the central  $\delta$ -term, which vanish on  $u$ , and the remaining correction vanishes because  $i + 2b > b$ , so  $f_{i+2b} u = 0$ . The other inducing relations are  $f_j w_1 = 0$  ( $j \geq b$ ),  $h_i w_1 = (\lambda_i + 2\delta_{i,0})w_1$  and  $K w_1 = \kappa w_1$ . Since  $Y u = 0$ ,  $L > 0$  and  $f_{a+1+2b} u = 0$ ,

$$Y w_m = -m\lambda_L w_{m+1}.$$

Use  $\mathcal{X}_{a,b} = \{e(\alpha)h(\beta)f(\gamma) : \alpha \in \Gamma_a, \beta \in \Gamma_{-1}, \gamma \in \Gamma_{b-1}\}$  and the source basis  $\{XY^p v\}$ . The derivation  $\text{ad } f_b$  strictly lowers  $d(X) = 2\ell(\alpha) + \ell(\beta)$ , so the same induction and highest-degree comparison apply with coefficients  $(-1)^p p! \lambda_L^p \neq 0$ .  $\square$

Iterating Corollary 2.5 (applied at successive boundaries, each step requiring the corresponding  $\lambda$  to be nonzero) yields  $\mathcal{T}_{f_c} \cdots \mathcal{T}_{f_b} M_{a,b} \cong M_{a+b-c+1,c-1}(\chi^{(b-c+1)})$  with  $\chi^{(k)}(h_0) = \lambda_0 + 2k$  and all other parameters fixed.

## 2.3 The derivation

Let  $\tilde{\mathfrak{g}} = \mathfrak{g} \rtimes \mathbb{C}d$  with  $[d, x_i] = ix_i$ , and  $\mathcal{I}(V) = U(\tilde{\mathfrak{g}}) \otimes_U V$ . Then  $\mathcal{I}(V) \cong \mathbb{C}[d] \otimes V$  with

$$x_i(P(d) \otimes v) = P(d - i) \otimes x_i v. \quad (11)$$

**Proposition 2.6.** *For  $F = f_c$  acting injectively on  $V$ ,*

$$\mathcal{T}_F \mathcal{I}(V) \cong \mathcal{I}(\mathcal{T}_F V), \quad F^{-m}(P(d) \otimes v) \mapsto P(d + mc) \otimes F^{-m}v.$$

*Proof.*  $\mathcal{I}$  is exact and faithful. It remains to check that the displayed map intertwines the action. For  $x = h_r$ :

$$h_r F^{-m} P(d) \otimes v = F^{-m} h_r P(d) \otimes v + 2m F^{-m-1} f_{r+c} P(d) \otimes v = F^{-m} P(d - r) \otimes h_r v + 2m F^{-m-1} P(d - (r+c)) \otimes f_{r+c} v,$$

which is sent to  $P(d - r + mc) \otimes F^{-m} h_r v + 2m P(d - (r+c) + c(m+1)) \otimes F^{-m-1} f_{r+c} v = P(d + mc - r) \otimes h_r F^{-m} v$ , the image of the right side. For  $x = e_r$  the same computation uses

$$e_r F^{-m} = F^{-m} e_r - m F^{-m-1} (h_{r+c} + r \delta_{r+c,0} K) - m(m+1) F^{-m-2} f_{r+2c},$$

and  $f_r$  commutes with  $F$ .  $\square$

### 3 Structure of the quotients $Q_c$ , $c < b$

Throughout this section  $c < b$ . Let  $\mathcal{X}$  be the set of ordered monomials  $X = f_{j_1} \cdots f_{j_p} e_{i_1} \cdots e_{i_q} h_{r_1} \cdots h_{r_s}$  with  $j_t \leq b$ ,  $j_t \neq c$ ,  $i_t \leq a$ ,  $r_u < 0$  (including the empty word), and put  $\deg X = 2q - 2p$ . For a nonzero  $v \in Q_c$  we denote by  $n_F(v)$  the smallest integer  $m \geq 1$  such that  $F^m v = 0$ ; in terms of the expansion  $v = \sum_{m \geq 1, X} c_{m,X} F^{-m} X u + M$ ,  $n_F(v)$  is the largest  $m$  with  $c_{m,X} \neq 0$  for some  $X$ .

**Proposition 3.1.** *The set  $\{F^{-m} X u + M : m \geq 1, X \in \mathcal{X}\}$  is a basis of  $Q_c$ , and*

$$\ker(F^r|_{Q_c}) = \text{span}_{\mathbb{C}}\{F^{-m} X u + M : 1 \leq m \leq r, X \in \mathcal{X}\}.$$

*The elements  $f_j$  act as follows:  $f_c$  surjectively and locally nilpotently;  $f_j$  injectively but not surjectively for  $j \leq b$ ,  $j \neq c$ ;  $f_j$  locally nilpotently for  $j > b$ . Moreover  $Q_c$  is a smooth weight module with  $K = \kappa$  and  $h_0$  semisimple,  $\text{Supp}_{h_0} Q_c = \lambda_0 + 2\mathbb{Z}$ , each nonzero weight space has countable dimension, and  $Q_c$  has no nonzero finite-dimensional submodules or quotients.*

*Proof.* The basis and the kernel formula follow from the PBW basis with  $F$  placed first and from  $M$  being free over  $\mathbb{C}[F]$ . For the elements  $f_j$ :  $F^r(F^{-m} X u) = F^{-m+r} X u$  vanishes for  $r \geq m$  and  $F^{-m} X u = F(F^{-m-1} X u)$ , giving surjectivity and local nilpotence of  $f_c$ . For  $c < j \leq b$ , the element  $f_j$  is a PBW basis element of the complement of  $S_{a,b}$  commuting with  $F$ , so  $f_j(F^{-m} X u) = F^{-m} f_j X u$  is multiplication by an independent variable; it is injective, and its image consists of vectors whose expansion is divisible by  $f_j$ , so  $w_1$  has no preimage and  $f_j$  is not surjective. For  $j > b$ :  $f_j u = 0$ ,  $f_j$  commutes with  $F$  and with all  $f_k$ , and the adjoint action of  $f_j$  raises indices, since  $[f_j, h_r] = 2f_{j+r}$  with  $j+r > b$  and  $[f_j, e_i] = -h_{j+i} - j\delta_{j+i,0}K$ ; hence iterating  $f_j$  on a PBW basis vector eventually produces only elements that annihilate  $u$ , and  $f_j$  is locally nilpotent. The weight formula is  $h_0(F^{-m} X u) = (\lambda_0 + 2m + \deg X)F^{-m} X u$ , and powers of  $h_{-1}$  produce infinitely many independent vectors in each weight space. Smoothness follows from (12) below: for a fixed PBW basis vector,  $r, r+c, r+2c$  large enough annihilate  $u$  and the central  $\delta$ -term vanishes. A finite-dimensional quotient cannot carry  $f_c$  both surjective and locally nilpotent, and a finite-dimensional smooth module is trivial, which is excluded because  $f_{c+1}$  is injective; this proves the last assertion.  $\square$

In the full localization, for all  $i \in \mathbb{Z}$  and  $m \geq 1$ ,

$$\begin{aligned} h_i F^{-m} &= F^{-m} h_i + 2m F^{-m-1} f_{i+c}, \\ e_i F^{-m} &= F^{-m} e_i - m F^{-m-1} (h_{i+c} + i\delta_{i+c,0}K) - m(m+1) F^{-m-2} f_{i+2c}, \end{aligned} \tag{12}$$

obtained from (6) with  $D = \text{ad } f_c$ :  $D(f_n) = 0$ ,  $D(h_n) = 2f_{n+c}$ ,  $D(e_n) = -(h_{n+c} + n\delta_{n+c,0}K)$ ,  $D^2(e_n) = -2f_{n+2c}$ ,  $D^3(e_n) = 0$ .

**Proposition 3.2.** *If  $1 \leq i \leq b-c$ , then for  $0 \neq v \in Q_c$ ,  $z \in \mathbb{C}$  and  $0 \neq p \in \mathbb{C}[t]$ ,*

$$n_F((h_i - z)v) = n_F(v) + 1, \quad p(h_i)v \neq 0.$$

*If  $i > b-c$ , then  $h_i - \lambda_i$  is locally nilpotent on  $Q_c$ . Hence  $Q_c$  is not isomorphic to any standard module  $M_{a',b'}(\psi)$ , and*

$$\text{Hom}_{\mathfrak{g}}(M_{a',b'}(\psi), Q_c) = 0, \quad \text{Hom}_{\mathfrak{g}}(\mathcal{T}_{f_c} M, \mathcal{T}_{f_{c'}} M) = 0 \quad (c \neq c', c, c' \leq b).$$

*Proof.* Write  $v = \sum_{m,X} c_{m,X} F^{-m} X u$  and let  $m_0 = n_F(v)$ . By (12),

$$h_i v = \sum_{m,X} c_{m,X} (F^{-m} h_i X u + 2m F^{-m-1} f_{i+c} X u).$$

Applying  $F^{m_0+1}$  kills every term, so  $F^{m_0+1}(h_i - z)v = 0$ . On the other hand, applying  $F^{m_0}$  kills all terms except the terms with  $m = m_0$  of the second sum, which contribute  $2m_0 F^{-1} f_{i+c} X_{m_0} u$ ; here  $c + i \leq b$  and  $c + i \neq c$ , so  $f_{i+c}$  is a PBW basis element of the complement of  $S_{a,b}$ , and this vector is nonzero in  $Q_c$ . Thus  $n_F((h_i - z)v) = m_0 + 1$ . Iterating,  $n_F(p(h_i)v) = m_0 + \deg p$ , so  $p(h_i)v \neq 0$ .

If  $i > b - c$ , then  $i + c > b$ , so  $h_i w_m = \lambda_i w_m$ , and for a PBW basis vector,  $(h_i - \lambda_i)^N X w_m = (\text{ad } h_i)^N(X) w_m$ . By the Leibniz rule, each term of  $(\text{ad } h_i)^N(X)$  is a product in which each factor is either unchanged, or has received  $k \geq 1$  adjoint actions of  $h_i$  and has index increased by  $ki$ , or is a scalar multiple of  $K$  arising from  $[h_i, h_{-i}]$ . An  $f$ -factor can receive at most  $\lfloor (b - j)/i \rfloor$  adjoint actions before its index exceeds  $b$  and it annihilates  $u$ ; an  $e$ -factor at most  $\lfloor (a - j)/i \rfloor$ ; an  $h$ -factor at most one (because  $[h_i, [h_i, h_{-i}]] = 0$  and  $[h_i, K] = 0$ ). Hence for  $N > \sum_t \lfloor (b - j_t)/i \rfloor + \sum_t \lfloor (a - j_t)/i \rfloor + s$  every term of  $(\text{ad } h_i)^N(X)$  either contains a factor annihilating  $u$  or vanishes, so  $(h_i - \lambda_i)^N X w_m = 0$ .

A standard module has a cyclic vector annihilated by the root vectors of its inducing subalgebra and which is an eigenvector of  $h_1$ ; the image of such a vector under a homomorphism into  $Q_c$  would be a nonzero vector on which  $h_1$  acts by a scalar, contradicting  $n_F((h_1 - \lambda)v) = n_F(v) + 1$  for  $\lambda \in \mathbb{C}$ . This gives the first Hom-vanishing. The second follows since  $f_c$  is locally nilpotent on  $\mathcal{T}_{f_c} M$  but injective on  $\mathcal{T}_{f_{c'}} M$  for  $c' \neq c$  (Proposition 3.1).  $\square$

**Proposition 3.3** (Commuting quotient localizations). *For  $j \leq b$ ,  $j \neq c$ ,*

$$\mathcal{T}_{f_j} \mathcal{T}_{f_c} M \cong \frac{D_{f_c, f_j} M}{D_{f_c} M + D_{f_j} M} \cong \mathcal{T}_{f_c} \mathcal{T}_{f_j} M.$$

*Proof.* From  $0 \rightarrow M \rightarrow D_{f_c} M \rightarrow \mathcal{T}_{f_c} M \rightarrow 0$  and the flatness of the Ore localization at  $f_j$ ,  $D_{f_j} \mathcal{T}_{f_c} M \cong D_{f_c, f_j} M / D_{f_j} M$ . Quotienting by the image of  $\mathcal{T}_{f_c} M$  corresponds to dividing by  $(D_{f_c} M + D_{f_j} M) / D_{f_j} M$ ; the displayed quotient is symmetric in  $f_c, f_j$ .  $\square$

**Proposition 3.4** (Annihilators, nonsplitting, derived functors). *In the ordinary enveloping algebra,*

$$\text{Ann}_U M = \text{Ann}_U D_F M = \text{Ann}_U Q_c,$$

*the sequence  $0 \rightarrow M \rightarrow D_F M \rightarrow Q_c \rightarrow 0$  does not split,  $\text{Hom}_{\mathfrak{g}}(Q_c, M) = \text{Hom}_{\mathfrak{g}}(Q_c, D_F M) = 0$ , and for  $\mathcal{T}_F V = (U_F/U) \otimes_U V$  one has*

$$L_1 \mathcal{T}_F(V) \cong \{v \in V : F^n v = 0 \text{ for some } n \geq 1\}, \quad L_j \mathcal{T}_F(V) = 0 \quad (j \geq 2).$$

*In particular  $D_F Q_c = \mathcal{T}_F Q_c = 0$  and  $L_1 \mathcal{T}_F(Q_c) \cong Q_c$ .*

*Proof.*  $\text{Ann}_U M$  is a two-sided ideal, hence stable under  $\Delta = \text{ad } F$ ; the commutation formula (6) then shows that it annihilates  $D_F M$ , and  $M \hookrightarrow D_F M$  gives  $\text{Ann}_U D_F M \subseteq \text{Ann}_U M \subseteq \text{Ann}_U Q_c$ . Conversely, if  $sQ_c = 0$  and  $\Delta^{t+1}(s) = 0$ , then for  $v \in M$  and  $m \geq 1$  we have  $sF^{-m}v \in M$ , and multiplying (6) by  $F^{m+t}$  gives

$$P_v(m) := \sum_{j=0}^t \binom{m+j-1}{j} F^{t-j} \Delta^j(s)v \in F^{m+t} M.$$

This is a vector-valued polynomial in  $m$  whose coefficients involve only finitely many PBW basis vectors, so their nonnegative  $F$ -degrees have a common upper bound; for  $m$  large the span meets  $F^{m+t} M$  trivially. Hence  $P_v(m) = 0$  for all large  $m$ , so the polynomial is identically zero; at  $m = 0$  only  $j = 0$  survives ( $\binom{j-1}{j} = 0$ ), giving  $F^t s v = 0$ , and  $F$  is injective on  $M$ . The splitting and Hom-vanishing follow from the  $F$ -torsion of  $Q_c$  versus the injectivity of  $F$  on  $M$  and  $D_F M$ . For the derived functors use the flat resolution  $0 \rightarrow U \rightarrow U_F \rightarrow U_F/U \rightarrow 0$  of the right  $U$ -module  $U_F/U$ :  $\mathcal{T}_F V = (U_F/U) \otimes_U V$  and  $L_1 \mathcal{T}_F V = \ker(V \rightarrow D_F V)$  is the  $F$ -torsion, while  $L_j = 0$  for  $j \geq 2$  by length one.  $\square$

## 4 The case $c \leq -a - 1$ : simplicity and the unique maximal submodule

Assume  $c \leq -a - 1$  and set

$$E = e_{-c}, \quad F = f_c, \quad H = h_0 - cK, \quad \mu = \lambda_0 - c\kappa. \quad (13)$$

Then  $[H, E] = 2E$ ,  $[H, F] = -2F$ ,  $[E, F] = H$ , so the subalgebra generated by  $E, H, F$  is isomorphic to  $\mathfrak{sl}_2$ ; the assumption  $c \leq -a - 1$  means  $-c > a$ , so  $Eu = 0$  and  $Hu = \mu u$ , and  $E$  is locally nilpotent on  $M$ . The identity

$$EF^{-m} = F^{-m}E - mF^{-m-1}H - m(m+1)F^{-m-2}F \quad (14)$$

gives, on  $w_m = F^{-m}u + M$ ,

$$Ew_m = -m(\mu + m + 1)w_{m+1}, \quad Hw_m = (\mu + 2m)w_m. \quad (15)$$

Define

$$G_c = \{v \in Q_c : E^n v = 0 \text{ for some } n \geq 1\}.$$

Since  $\text{ad } E$  is locally nilpotent,  $G_c$  is a  $\mathfrak{g}$ -submodule: for  $v \in G_c$  and  $s \in U$ ,  $E^r(sv) = \sum_{j=0}^r \binom{r}{j} (\text{ad } E)^j(s) E^{r-j}v$ , which vanishes for  $r$  large. Every nonzero submodule of  $G_c$  contains a nonzero vector of  $P_\nu := \ker E \cap G_c \cap (Q_c)_\nu$  for some  $\nu$ : take the last nonzero  $E$ -image.

**Proposition 4.1.** *Without assuming (2),  $G_c$  is a proper submodule of  $Q_c$  and*

$$G_c = 0 \iff \mu \notin \mathbb{Z}.$$

If  $\mu \in \mathbb{Z}$  and  $k \geq \max(1, \mu + 2)$ , then  $0 \neq f_b^k w_1 \in G_c$  and  $E^{2k-\mu-1}(f_b^k w_1) = 0$ . Moreover  $Q_c$  is cyclic,  $Q_c = Uw_1$  if and only if  $\mu \notin \{-2, -3, \dots\}$ , and for  $\mu = -s$ ,  $s \geq 2$ , one has  $Q_c = Uw_s \neq Uw_1$ . The operator  $E$  is injective on  $Q_c/G_c$ .

*Proof.* By (15),  $w_m \notin G_c$  for  $m$  large (all coefficients  $-m(\mu + m + 1)$  are nonzero from some point on), so  $G_c$  is proper. If  $0 \neq v \in \ker E$  is an  $H$ -weight vector of weight  $\nu$  and  $n$  is minimal with  $F^n v = 0$ , then

$$0 = EF^n v = \sum_{i=0}^{n-1} F^i [E, F] F^{n-1-i} v = \sum_{i=0}^{n-1} (\nu - 2(n-1-i)) F^{n-1} v = n(\nu - n + 1) F^{n-1} v,$$

so  $\nu = n - 1 \in \mathbb{Z}$ ; but  $\nu \in \mu + 2\mathbb{Z}$ , whence  $G_c = 0$  for  $\mu \notin \mathbb{Z}$  (every nonzero vector of  $G_c$  has a nonzero weight component, and its last nonzero  $E$ -image lies in  $\ker E$ ).

For  $\mu \in \mathbb{Z}$ , put  $Y = f_b$  and  $\eta = \lambda_{b-c}$ . Since  $[E, Y] = h_{b-c}$  (the  $\delta$ -term vanishes because  $b-c \geq 1$ ) and  $[H, Y] = -2Y$ , while  $[h_{b-c}, Y] = -2f_{b+(b-c)}$  annihilates  $u$ , we get the commutation formula on  $\mathbb{C}[F, Y]u$ :

$$E(F^j Y^\ell u) = j(\mu - j + 1 - 2\ell) F^{j-1} Y^\ell u + \ell \eta F^j Y^{\ell-1} u. \quad (16)$$

Indeed  $EF^j = F^j E + jF^{j-1}(H - j + 1)$  and  $[E, Y^\ell]u = \ell Y^{\ell-1} h_{b-c} u = \ell \eta Y^{\ell-1} u$ . Define  $v_k = \sum_{j=0}^k a_j F^j Y^{k-j} u$  with

$$a_0 = 1, \quad a_j = -\frac{(k-j+1)\eta}{j(\mu - 2k + j + 1)} a_{j-1}. \quad (17)$$

The denominators are nonzero because  $\mu - 2k + j + 1 \leq \mu - k + 1 < 0$ . In (16) the coefficient of  $F^{j-1} Y^{k-j} u$  in  $Ev_k$  is  $j(\mu - 2k + j + 1)a_j + (k-j+1)\eta a_{j-1} = 0$ , so  $Ev_k = 0$  and  $Hv_k = (\mu - 2k)v_k$ . Set  $\nu = \mu - 2k \leq -2$ . By (14),  $E(F^{-m}v_k) = -m(\nu + m + 1)F^{-m-1}v_k$  for all  $m \geq 1$ ; iterating,

$$E^r(F^{-1}v_k) = \left( \prod_{t=1}^r (-t)(\nu + t + 1) \right) F^{-1-r}v_k,$$

and the factor with  $t = -\nu - 1$  vanishes, so  $E^{-\nu-1}(F^{-1}v_k) = 0$ . Passing to the quotient,  $F^{-1}v_k + M = Y^k w_1 \neq 0$  (the terms with  $j \geq 1$  land in  $M$ ), giving  $0 \neq Y^k w_1 \in G_c$  and  $E^{2k-\mu-1}(Y^k w_1) = 0$ .

The module is cyclic: (15) produces  $w_{m+1}$  from  $w_m$  whenever  $-m(\mu + m + 1) \neq 0$ , and  $Fw_m = w_{m-1}$  ( $w_0 = 0$ ) produces the lower ones; to pass from the  $w_m$  to arbitrary PBW basis vectors, grade the words by the weight  $\rho = 2\#e + \#h$ : commuting  $X$  past  $F^{-m}$  gives  $Xw_m = F^{-m}Xu + M + (\text{terms of strictly lower } \rho)$ , and induction on  $\rho$ , simultaneously for all  $m$ , proves that the vectors  $w_m$  generate  $Q_c$ . For  $\mu = -s$ ,  $s \geq 2$ , the coefficient in (15) vanishes at  $m = s - 1$ , so  $w_1 \in G_c$  and  $Uw_1 \subseteq G_c \neq Q_c$ , while all coefficients from  $w_s$  upward are nonzero, so  $Q_c = Uw_s$ . Finally, if  $Ev \in G_c$  then  $v \in G_c$ , so  $E$  is injective on  $Q_c/G_c$ .  $\square$

#### 4.1 The containment of all proper submodules

We isolate the abstract mechanism. Let  $\mathfrak{a}$  be a complex Lie algebra containing elements  $E, H, F$  with  $[H, E] = 2E$ ,  $[H, F] = -2F$ ,  $[E, F] = H$ ; write  $A = U(\mathfrak{a})$  and assume:

- (i)  $\text{ad } E$  and  $\text{ad } F$  are locally nilpotent on  $A$ , and the Ore localization  $A_F$  at the powers of  $F$  exists;
- (ii)  $M$  is a simple  $A$ -module on which  $H$  acts semisimply,  $E$  acts locally nilpotently, and  $F$  acts injectively.

Let  $D = A_F \otimes_A M$ ,  $Q = D/M$ , and  $G = \{q \in Q : E^n q = 0 \text{ for some } n \geq 1\}$ . The  $H$ -action on  $D$  and  $Q$  is semisimple since  $[H, F^{-m}] = 2mF^{-m}$ , and  $G$  is an  $A$ -submodule by the Leibniz expansion. Define the Casimir element of the subalgebra generated by  $E, H, F$ :

$$C = H^2 + 2H + 4FE. \quad (18)$$

**Lemma 4.2.**  $C$  commutes with  $E, H, F$ . For  $H z = \nu z$  and  $n \geq 0$ ,

$$F^n E^n z = A_n(C; \nu) z, \quad A_n(t; \nu) = 4^{-n} \prod_{j=0}^{n-1} (t - (\nu + 2j)(\nu + 2j + 2)), \quad (19)$$

with  $A_0 = 1$ . Consequently  $C$  acts locally finitely on  $M$  and on  $D$ .

*Proof.*  $[C, E] = [H^2, E] + 2[H, E] + 4[FE, E] = 4EH + 4E + 4E - 4HE = 0$ , since  $[H^2, E] = H[H, E] + [H, E]H = 2HE + 2EH = 4EH + 4E$  and  $[F, E] = -H$ ; similarly  $[C, F] = 0$ . For  $n = 1$ ,  $FEz = \frac{1}{4}(C - H^2 - 2H)z = \frac{1}{4}(C - (\nu^2 + 2\nu))z = A_1(C; \nu)z$ . If (19) holds for  $n$ , apply it to the weight vector  $Ez$  (weight  $\nu + 2$ ) and multiply by  $F$  on the left:  $F^{n+1}E^{n+1}z = F \cdot A_n(C; \nu + 2)Ez = A_n(C; \nu + 2)FEz = A_{n+1}(C; \nu)z$ . If  $z \in M$  is a weight vector with  $E^n z = 0$ , then  $A_n(C; \nu)z = F^n E^n z = 0$  is a nonzero polynomial in  $C$  annihilating  $z$ ; each vector of  $M$  has finitely many weight components, so a product of such polynomials kills it. As  $C$  commutes with  $F^{-1}$ , the same polynomials work in  $D$ .  $\square$

For an  $A$ -submodule  $P \subseteq D$  define

$$W(P) = \bigcap_{n \geq 0} F^n P. \quad (20)$$

**Lemma 4.3.**  $W(P)$  is an  $A$ -submodule of  $D$ , is stable under  $F^{-1}$ , and if  $W(P) \neq 0$  then  $P = D$ .

*Proof.* For  $s \in A$  choose  $\ell$  with  $(\text{ad } F)^{\ell+1}(s) = 0$ . We use the binomial identity

$$sF^{n+\ell} = \sum_{j=0}^{\ell} (-1)^j \binom{n+\ell}{j} F^{n+\ell-j} (\text{ad } F)^j(s), \quad (21)$$

which holds in  $A$ : for  $N = 1$  it reads  $sF = Fs - [F, s]$ , and the induction step is  $sF^{N+1} = \sum_j (-1)^j \binom{N}{j} F^{N-j} (\text{ad } F)^j(s) F = \sum_j (-1)^j \binom{N}{j} F^{N-j+1} (\text{ad } F)^j(s) + \sum_{j \geq 1} (-1)^j \binom{N}{j-1} F^{N-j+1} (\text{ad } F)^j(s)$ , whose coefficients  $\binom{N}{j} + \binom{N}{j-1} = \binom{N+1}{j}$  give the identity for  $N+1$ . If  $w \in W(P)$ , write  $w = F^{n+\ell} p$  with  $p \in P$ ; then  $sw = \sum_j (-1)^j \binom{n+\ell}{j} F^{n+\ell-j} (\text{ad } F)^j(s) p \in F^n P$  for every  $n$ , so  $sw \in W(P)$ . Writing  $w = F^{n+1} p$  gives  $F^{-1}w = F^n p \in W(P)$ , so  $W(P)$  is  $F^{-1}$ -stable. If  $0 \neq w \in W(P)$ , write  $w = F^{-m} z$  with  $z \in M$ ; then  $F^m w = z \in M \cap W(P)$  is nonzero ( $F$  is invertible on  $D$ ). Since  $M$  is simple,  $M \subseteq W(P)$ , and  $F^{-1}$ -stability gives  $D = A_F M \subseteq W(P) \subseteq P$ , so  $P = D$ .  $\square$

**Theorem 4.4** (Containment of all proper submodules). *Under (i)–(ii), every proper  $A$ -submodule  $N \subsetneq Q$  is contained in  $G$ . If  $G \neq Q$ , then  $G$  is the unique maximal submodule of  $Q$ ,  $Q/G$  is simple,  $Q$  is indecomposable, and if  $G \neq 0$  the sequence  $0 \rightarrow G \rightarrow Q \rightarrow Q/G \rightarrow 0$  does not split.*

*Proof.* Let  $N \subseteq Q$  be a submodule not contained in  $G$ , and let  $P \subseteq D$  be its inverse image, so  $M \subseteq P$  and  $P/M = N$ . Choose  $q \in N$  on which  $E$  is not locally nilpotent; since  $H$  is semisimple and each vector has finitely many weight components, one weight component  $q_0$  of  $q$  has the same property, and  $q_0 \in N$  (it is obtained from  $q$  by a polynomial in  $H$ ). Lift  $q_0$  to a weight vector  $w \in P$ ,  $Hw = \eta w$ . By Lemma 4.2 choose  $0 \neq p(t)$  with  $p(C)w = 0$ . The set of nonnegative integers  $k$  with  $p((\eta + 2k)(\eta + 2k + 2)) = 0$  is finite, each root of  $p$  contributing at most two values of  $k$ ; choose  $r$  larger than all of them and set  $v = E^r w$ ,  $\nu = \eta + 2r$ . Then  $E^r q_0 \neq 0$ , so  $v \neq 0$ ; also  $Hv = \nu v$ ,  $p(C)v = 0$  and  $p((\nu + 2j)(\nu + 2j + 2)) \neq 0$  for all  $j \geq 0$ .

For each  $n \geq 1$ , the polynomials  $p(t)$  and  $A_n(t; \nu)$  are therefore coprime, so there are  $B_n, R_n \in \mathbb{C}[t]$  with

$$A_n(t; \nu) B_n(t) + p(t) R_n(t) = 1.$$

The vector  $B_n(C)v$  has weight  $\nu$ , and (19) gives

$$F^n(E^n B_n(C)v) = A_n(C; \nu) B_n(C)v = v,$$

so  $v \in F^n P$  for every  $n \geq 0$ :  $0 \neq v \in W(P)$ . Lemma 4.3 gives  $P = D$ , hence  $N = Q$ . The remaining assertions follow: if  $G \neq Q$  it is a proper submodule containing every proper submodule; a nontrivial direct sum decomposition of  $Q$  would put both summands in  $G$ ; a splitting of the displayed sequence would give such a decomposition when  $G \neq 0$ .  $\square$

**Theorem 4.5** (Simplicity criterion and the simple quotient). *Assume*

$$\lambda_L \neq 0, \quad \kappa \neq -2, \quad c < b, \quad c \leq -a - 1. \quad (22)$$

*Then*

$$Q_c \text{ is simple} \iff \mu = \lambda_0 - c\kappa \notin \mathbb{Z}.$$

*For every  $\mu$ ,  $G_c$  is the unique maximal submodule of  $Q_c$ , the quotient  $Q_c/G_c$  is simple, and  $Q_c$  is indecomposable; for  $\mu \in \mathbb{Z}$  the sequence  $0 \rightarrow G_c \rightarrow Q_c \rightarrow Q_c/G_c \rightarrow 0$  does not split.*

*Proof.* By (2),  $M$  is simple. The PBW basis makes  $F$  injective and  $H$  semisimple on  $M$ ;  $E = e_{-c}$  is locally nilpotent on  $M$  because  $-c > a$ ; and  $\text{ad } E, \text{ad } F$  are locally nilpotent on  $U$ : indeed  $[e_r, e_s] = [f_r, f_s] = 0$  and, by (1),  $[e_r, [e_r, [e_r, x]]] = [f_r, [f_r, [f_r, x]]] = 0$  for every generator  $x$ , the chains being  $f_s \mapsto h_{r+s} \mapsto e_{2r+s} \mapsto 0$  and the analogous one for  $f_r$ . Thus Theorem 4.4 applies: every proper submodule of  $Q_c$  is contained in  $G_c$ . By Proposition 4.1,  $G_c = 0$  exactly when  $\mu \notin \mathbb{Z}$ , and  $0 \neq G_c \neq Q_c$  when  $\mu \in \mathbb{Z}$ . The criterion and the structural assertions follow.  $\square$

## 5 Central reduction and the derivation

**Proposition 5.1** (Reduction by commuting endomorphisms). *Let  $V$  be a  $U$ -module, let  $U_F$  be an Ore localization flat as a right  $U$ -module, and let a commutative algebra  $R$  act on  $V$  by  $U$ -module endomorphisms. The action extends to  $D_F V$  by  $z(s \otimes v) = s \otimes zv$  and descends to  $\mathcal{T}_F V$ . For any algebraic ideal  $I \subseteq R$ ,*

$$D_F(V/IV) \cong D_F V / I(D_F V), \quad \boxed{\mathcal{T}_F(V/IV) \cong (\mathcal{T}_F V) / I(\mathcal{T}_F V),}$$

and injectivity of  $F$  on  $V/IV$  is not required.

*Proof.* The extension is well defined because each  $z$  commutes with the  $U$ -action, hence with  $F^{-1}$ :  $F^{-m}(zv) = z(F^{-m}v)$ . Localizing  $0 \rightarrow IV \rightarrow V \rightarrow V/IV \rightarrow 0$  and using flatness gives  $D_F(V/IV) \cong D_F V / \text{Im } D_F(IV)$ ; the image equals  $I(D_F V)$ , since every element is a finite sum of terms  $F^{-m}(zv) = z(F^{-m}v)$ . The image of  $V/IV$  in  $D_F V / I(D_F V)$  is  $(\ell(V) + I(D_F V)) / I(D_F V)$ , whose cokernel is  $D_F V / (\ell(V) + I(D_F V))$ ; quotienting  $\mathcal{T}_F V = D_F V / \ell(V)$  by  $I(\mathcal{T}_F V)$  produces exactly the same expression.  $\square$

At the critical level  $\kappa = -2$  the Segal–Sugawara operators are the normally ordered sums

$$T(n) = \frac{1}{4} \sum_{j \in \mathbb{Z}} (2 : e_{-j} f_{n+j} : + 2 : f_{-j} e_{n+j} : + : h_{-j} h_{n+j} :), \quad (23)$$

where  $: x_p y_q := x_p y_q$  if  $p < 0$  and  $: y_q x_p := y_q x_p$  if  $p \geq 0$ . On a smooth module each sum is finite, and at  $\kappa = -2$  the  $T(n)$  commute with  $\mathfrak{g}$  [1, Section 2.5].

**Lemma 5.2.** *Let  $V$  be a smooth module of level  $-2$  and  $F = f_c$ . The natural extension of the  $U$ -endomorphism  $T(n)$  from  $V$  to  $D_F V$  agrees with the actual Sugawara action there; in particular  $T(n)(F^{-m}v) = F^{-m}T(n)v$ .*

*Proof.* Localization preserves smoothness: for  $m \geq 1$ ,

$$h_r F^{-m} = F^{-m} h_r + 2m F^{-m-1} f_{r+c}, \quad e_r F^{-m} = F^{-m} e_r - m F^{-m-1} (h_{r+c} + r \delta_{r+c,0} K) - m(m+1) F^{-m-2} f_{r+2c},$$

and the  $f_r$  commute with  $F$ ; for a fixed PBW basis vector choose  $r, r+c, r+2c$  large enough that all resulting elements annihilate the vector and the central  $\delta$ -term vanishes. The map  $V \rightarrow D_F V$  intertwines  $T(n)$  (each finite product intertwines, and normal ordering makes the sum finite on each vector). Since  $T(n)$  commutes with  $\mathfrak{g}$  it commutes with  $F$ , hence with  $F^{-1}$ , which gives the displayed formula.  $\square$

**Corollary 5.3.** *Let  $V$  be a smooth module of level  $-2$ , fix integers  $n_i$  and scalars  $\theta_i$  ( $i \in J$ ), and let  $R = \mathbb{C}[z_i : i \in J]$  act by  $z_i = T(n_i)$  and  $I = (z_i - \theta_i : i \in J)$ . For every  $c \in \mathbb{Z}$ ,*

$$\mathcal{T}_{f_c}(V/IV) \cong (\mathcal{T}_{f_c} V) / I(\mathcal{T}_{f_c} V),$$

and the same holds for any algebraic ideal of  $R$ .

*Proof.* Apply Proposition 5.1 and Lemma 5.2.  $\square$

## 6 The subspace $G_c$ : established facts and open questions

Assume  $c \leq -a - 1$ . Applying the automorphism  $\sigma_{-c}$  of  $\mathfrak{g}$  defined by  $e_i \mapsto e_{i-c}$ ,  $f_i \mapsto f_{i+c}$ ,  $h_i \mapsto h_i - c \delta_{i,0} K$ ,  $K \mapsto K$ , we may assume  $c = 0$ , so that  $E = e_0$ ,  $F = f_0$ ,  $H = h_0$ , and  $Q_c$  is the quotient of  $M_{a+c, b-c}(\chi')$  at  $f_0$ , with  $\chi'(h_0) = \mu$ . We work in these coordinates in this section.

Let  $V_0 \subset M$  be the span of the PBW basis vectors not involving  $f_0$ , so  $M = \bigoplus_{r \geq 0} f_0^r V_0$ . For  $x \in V_0$  define linear maps  $A_r : V_0 \rightarrow V_0$  by the finite expansion

$$Ex = \sum_{r \geq 0} f_0^r A_r x, \quad A_0 = \pi_0 \circ \text{ad } E, \quad (24)$$

where  $\pi_0$  projects onto  $V_0$ . The following replaces the incorrect formula of an earlier draft; it was corrected after the independent audit of 2026-09-22, whose counterexamples are reproduced in `work/check_audit_regressions.py`.

**Lemma 6.1** (Corrected commutation formula for  $E$ ). *If  $x \in V_0$  with  $Hx = (\mu + \delta)x$  and  $m \geq 1$ , then in  $Q_c$ ,*

$$E(F^{-m}x) = \sum_{r=0}^{m-1} F^{-(m-r)} A_r x - m(\mu + \delta + m + 1)F^{-m-1}x. \quad (25)$$

*Proof.* By (14) and (24),

$$EF^{-m}x = F^{-m}Ex - mF^{-m-1}Hx - m(m+1)F^{-m-2}Fx = \sum_{r \geq 0} F^{-m}F^r A_r x - m(\mu + \delta)F^{-m-1}x - m(m+1)F^{-m-1}x;$$

the terms with  $r < m$  survive as  $F^{-(m-r)}A_r x$ , and the terms with  $r \geq m$  lie in  $M$  and vanish in  $Q_c$ .  $\square$

**Lemma 6.2.** *If  $0 \neq v \in Q_\nu$  and  $Ev = 0$ , then  $\nu \in \mathbb{Z}_{\geq 0}$ , and in the expansion  $v = \sum_{m=1}^N F^{-m}x_m$  with  $x_m \in V_0$  and  $x_N \neq 0$  one has  $N = \nu + 1$ .*

*Proof.* The coefficient of  $F^{-(N+1)}$  in  $Ev$  receives contributions only from the diagonal term  $-N(\mu + \delta_N + N + 1)x_N$  of (25), where  $\mu + \delta_N = \nu - 2N$ ; the sum  $\sum_r F^{-(m-r)}A_r x_m$  has negative powers at most  $m \leq N$ . Thus  $0 = -N(\nu - N + 1)x_N$  forces  $\nu = N - 1$ .  $\square$

**Proposition 6.3** (Triangular system for the vectors annihilated by  $E$ ). *A nonzero vector  $v = \sum_{m=1}^{\nu+1} F^{-m}x_m \in Q_\nu$  satisfies  $Ev = 0$  if and only if*

$$\sum_{m=k}^{\nu+1} A_{m-k}x_m = (k-1)(\nu - k + 2)x_{k-1}, \quad 1 \leq k \leq \nu + 1, \quad x_0 = 0. \quad (26)$$

For  $\nu = 1$  this reads  $x_1 = A_0x_2$  and  $(A_0^2 + A_1)x_2 = 0$ . The earlier description of the primitive space by the single condition  $A_0^{\nu+1}x_{\nu+1} = 0$  is false: the element  $Z = h_{-1}^2 + 4f_{-1}e_{-1} + 2h_{-2}$  commutes with  $E, F, H$ , and at  $\mu = -1$  the vector  $p = \lambda_1 F^{-1}u + F^{-2}f_1u$  lies in  $\ker E$  (weight 1) while  $Zp$  has layers  $x_1 = (\lambda_1 Z - 4h_{-1})u$ ,  $x_2 = (f_1 Z - 4(\kappa + 1)f_{-1})u$  with  $A_0^2x_2 = 8e_{-1}u \neq 0$  and  $A_1x_2 = -8e_{-1}u$ , so that (26) holds with  $k = 1$  by cancellation.

*Proof.* Equate the coefficient of  $F^{-k}$  in  $Ev = 0$  using (25): from  $\sum_r F^{-(m-r)}A_r x_m$  with  $m-r = k$  (i.e.  $r = m - k \geq 0$ ) we get  $\sum_{m \geq k} A_{m-k}x_m$ , and the diagonal contributes  $-(k-1)(\mu + \delta_{k-1} + k)x_{k-1} = -(k-1)(\nu - k + 2)x_{k-1}$ , using  $\mu + \delta_{k-1} = \nu - 2(k-1)$ . The assertions about  $Z$  are direct bracket computations ( $[E, Z] = [F, Z] = [H, Z] = 0$ ; the displayed values are verified in `work/check_audit_regressions.py`).  $\square$

**Proposition 6.4** (Decomposition into irreducibles of the subalgebra generated by  $E, H, F$ ). *Let  $\mathfrak{s}$  be the subalgebra generated by  $E, H, F$ , isomorphic to  $\mathfrak{sl}_2$ . As an  $\mathfrak{s}$ -module,*

$$G_c \cong \bigoplus_{\nu \in \mathbb{Z}_{\geq 0}} P_\nu \otimes L(\nu),$$

where  $L(\nu)$  is the  $(\nu + 1)$ -dimensional irreducible module and  $P_\nu = \ker E \cap G_c \cap (Q_c)_\nu$  is the multiplicity space (possibly infinite-dimensional). This is not a decomposition into  $\mathfrak{g}$ -submodules.

*Proof.*  $E$  and  $F$  are locally nilpotent on  $G_c$ , so every vector lies in the finite span  $\text{span}_{\mathbb{C}}\{F^r E^s v\}$ ; complete reducibility of the finite dimensional modules of  $\mathfrak{sl}_2$  gives the decomposition. A vector  $p \in \ker E$  with  $F^N p = 0$  generates  $L(\nu)$  with  $\nu$  its  $H$ -weight, and  $F^N p = 0$  forces  $\nu \leq N - 1$ , so  $\nu \in \mathbb{Z}_{\geq 0}$ .  $\square$

**Proposition 6.5** (A projection lowering the highest weight by 2). *For  $\nu \geq 2$ ,  $p \in P_\nu$  and  $j \in \mathbb{Z}$ ,*

$$\ell_j(p) := f_j p - \frac{1}{\nu} F h_j p - \frac{1}{\nu(\nu+1)} F^2 e_j p \in P_{\nu-2}. \quad (27)$$

*It is not identically zero: at  $\mu = -2$ , for  $p_2 = e_{-1} w_1 \in P_2$ ,*

$$\ell_1(p_2) = F^{-1} \left( \frac{1}{3} f_1 e_{-1} + \frac{\lambda_1}{6} h_{-1} + \frac{2\kappa-2}{3} \right) u + M \neq 0.$$

*Proof.* Compute  $E \ell_j(p)$  term by term.  $E f_j p = [E, f_j] p + f_j E p = h_j p$ . Since  $H h_j p = \nu h_j p$  and  $E h_j p = [E, h_j] p + h_j E p = -2 e_j p$ ,

$$E F h_j p = [E, F] h_j p + F E h_j p = H h_j p - 2 F e_j p = \nu h_j p - 2 F e_j p.$$

Since  $e_j p \in P_{\nu+2}$  (as  $[E, e_j] = 0$ ), we have  $E F e_j p = (\nu + 2) e_j p$  and

$$E F^2 e_j p = [E, F] F e_j p + F E F e_j p = H F e_j p + (\nu + 2) F e_j p = 2(\nu + 1) F e_j p,$$

because  $H F e_j p = F H e_j p - 2 F e_j p = (\nu + 2 - 2) F e_j p = \nu F e_j p$ . Hence

$$E \ell_j(p) = h_j p - \frac{1}{\nu} (\nu h_j p - 2 F e_j p) - \frac{1}{\nu(\nu+1)} 2(\nu + 1) F e_j p = 0,$$

and all three terms of (27) have  $H$ -weight  $\nu - 2$ . The displayed value of  $\ell_1(p_2)$  is obtained by the direct PBW computation reproduced in `work/check_audit_regressions.py`.  $\square$

**Remark 6.6** (Withdrawn claims and what is open). *An earlier draft claimed that the subspaces  $G^{\geq \nu} := \bigoplus_{\nu' \geq \nu} P_{\nu'} \otimes L(\nu')$  are  $\mathfrak{g}$ -submodules and deduced that  $G_c$  is never simple, that  $\text{soc}(G_c) = \text{soc}(Q_c) = 0$ , and that  $Q_c$  has infinite length. These claims are withdrawn. Counterexample: at  $\mu = -2$  and  $\kappa \neq 0$ , for  $p = w_1 \in L(0)$  we have  $h_{\pm 1} p = -F e_{\pm 1} p \in G^{\geq 2}$  (indeed  $h_r p = e_r F p - F e_r p$  and  $F p = 0$ , while  $e_{\pm 1} p \in P_2$ ); if  $G^{\geq 2}$  were a submodule,  $[h_1, h_{-1}] p = 2\kappa p \in G^{\geq 2}$ , contradicting  $p \in L(0)$ . Moreover the projection of Proposition 6.5 maps  $P_2$  into  $P_0$ , so these subspaces are not stable even at  $\kappa = 0$ . In the case  $\mu \in \mathbb{Z}$  the containment theorem still gives  $\text{soc}(Q_c) = \text{soc}(G_c)$  (every simple submodule of  $Q_c$  is proper and hence contained in  $G_c$ ), but the value of this socle, the simplicity of  $G_c$ , and the composition length of  $Q_c$  are open.*

## 7 Open problems

- Simplicity in the case  $-a \leq c < b$ .** Here the element  $e_{-c}$  is injective on  $Q_c$  (it is a PBW basis element of the complement of  $S_{a,b}$ ; grading by the weight  $2\#e + \#h$ , its action on the leading term is multiplication by an independent variable, and the commutation corrections have lower weight), so  $G_c = 0$ ; moreover  $Q_c = U w_1$  when  $\lambda_L \neq 0$ , by  $[e_{L-c}, f_c] = h_L$  and  $e_{L-c} w_m = -m \lambda_L w_{m+1}$ . The proof of Theorem 4.5 does not apply because  $e_{-c}$  is not locally nilpotent on  $M$ . The simplicity criterion for  $Q_c$  is unknown.
- Submodules of  $G_c$ .** For  $\mu \in \mathbb{Z}$ , determine whether  $G_c$  is simple, compute  $\text{soc}(G_c) = \text{soc}(Q_c)$ , and decide whether  $Q_c$  has finite composition length; a correct analysis must track the three channels  $P_\nu \rightarrow P_{\nu+2}, P_\nu, P_{\nu-2}$  simultaneously (Propositions 6.3 and 6.5).
- Critical reduction.** Corollary 5.3 gives  $\mathcal{T}_f(V/IV) \cong (\mathcal{T}_f V)/I(\mathcal{T}_f V)$ , but the simplicity of the reduced quotients at  $\kappa = -2$  is not determined.
- Parameter classification.** A complete isomorphism classification of the  $Q_c$ , including  $\lambda_L = 0$  and  $\kappa = -2$ , is open.

## References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024). <https://arxiv.org/abs/2404.03855v2>.
- [2] Project note, *Boundary and Nonboundary Localization Quotients of Smooth Affine Modules*, September 2026 (unified note and its addendum of September 15, 2026).
- [3] Independent audit of 2026-09-22, `audits/2026-09-22/audit_report_cn.md` and `audits/2026-09-22/independent_pbw_audit.py` (154 exact checks); counterexamples reproduced in `work/check_audit_regressions.py` (24 checks).